

The Wonderful World Of INGENIOUS PRINTED SENSORS

The electronics industry has heavily relied on sensors that have rigid proportions. But can the rise of flexible, printed sensors become a game-changer for a myriad of applications, where form-factor and convenience play the core role to implement a particular technology?



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Conventional electronics based on PCBs are produced efficiently, resulting in a very high yield. However, they are rigid, take up a lot of space, possess a fixed form-factor, and are not always lightweight. Although sensors such as thermistors can be small in terms of size, the majority are large and rigid. On the other hand, printed sensors are very thin and flexible. Also, their large-scale, cost-effective production allows rapid prototyping and customised printing for different applications.

So, what trends drive the adoption of printed, flexible sensors? A large part of this answer lies in the revolution brought forth by Industrial IoT (IIoT), or Industry 4.0. Here real-time monitoring can be performed digitally by retrofitting such sensors to industrial equipment for predictive maintenance.

Healthcare is the next area that drives their adoption, where electronic

skin patches offer the benefit of remote patients' health monitoring. Following it, smart homes and cities are the next segment that could drive this adoption. The idea of having several connected low-cost sensors around our lives is quite appealing and valuable for long-term use.

In a typical workplace, printed sensors would be able to check room temperature, monitor humidity, measure air-quality, and help in preventive maintenance. They do so by identifying vibrations, force-sensing via trackpads, printed fingerprint sensing, leak detection, and many more. Fig. 2 paints a vivid picture of this. However, not every sensor performs the same function as the other.

Types of sensors

Besides automation, electric pianos and medical applications, pressure sensors (piezoresistive or piezoelectric) have a scope for smart flooring and force-sensitive touchpads on laptops. Till date, temperature sensors were made with inorganic materials, but with printed flexible technology they can be done in 2D at a relatively cost-effective rate.

Optical (or light) printed sensors have a similar architecture to organic photovoltaics but are optimised for light detection rather than power generation. When it comes to e-textiles, capacitive strain sensors are an ideal choice as they

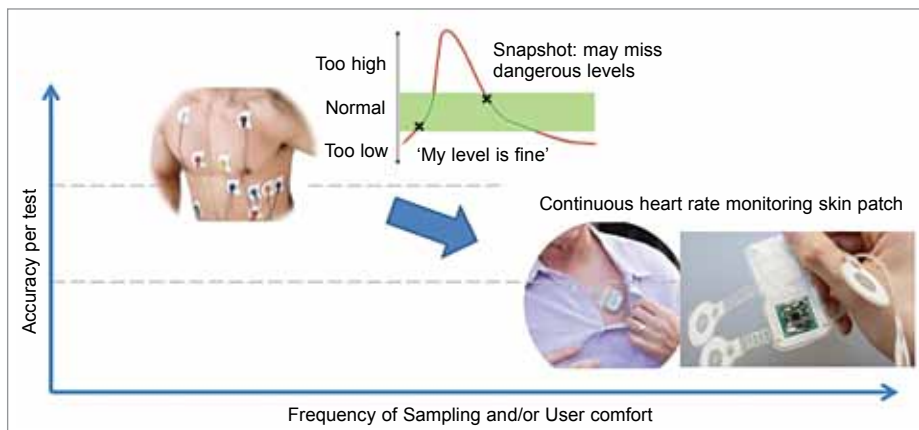


Fig. 1: Electronic skin patch for healthcare applications (Credit: IDTechEx)